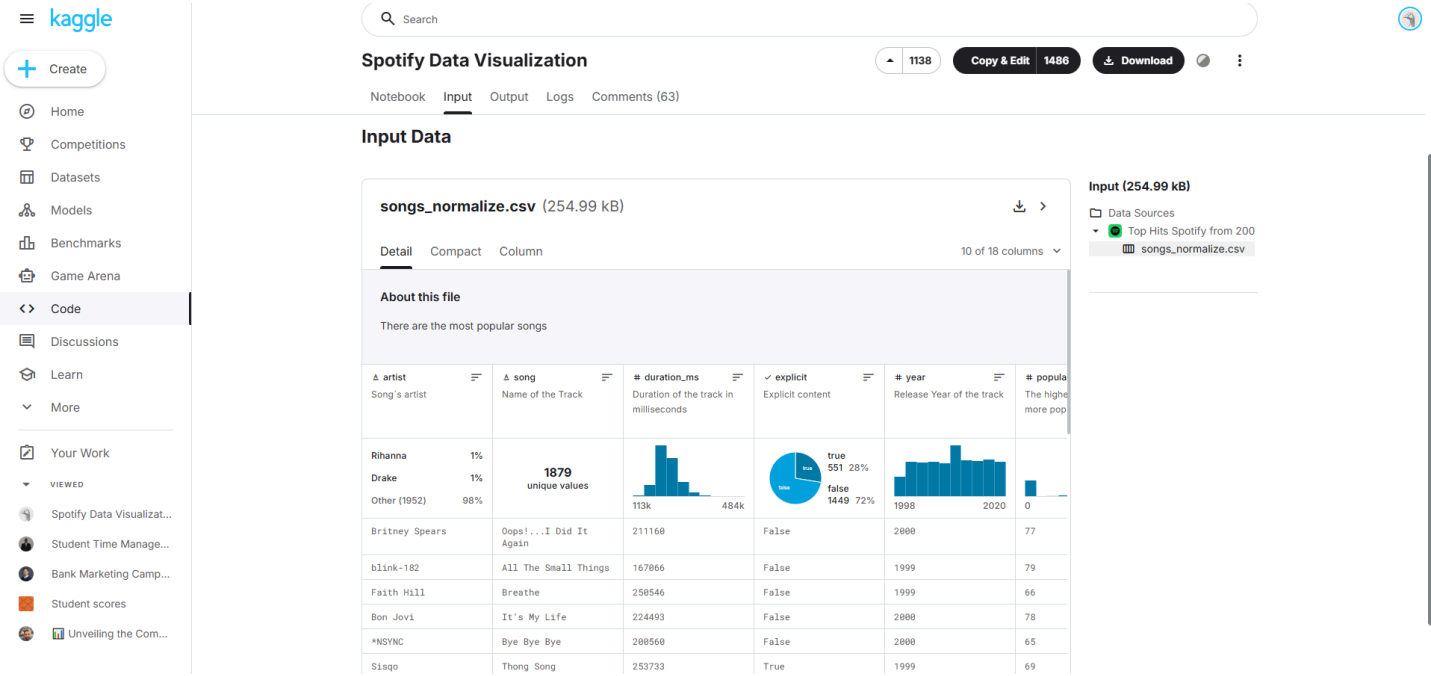




Predicting Song Popularity: A Machine Learning Study on Spotify Hits by John Michael Angelo C. Binonggo





I. SPOTIFY DATA VISUALIZATION AND HIT PREDICTION

Spotify has revolutionized music listening since its launch in Sweden in 2008, becoming the leading global audio streaming platform. With millions of users and a significant influence on the music industry, understanding what makes a song a hit is both intriguing and valuable. In this project, I explored the Top Hits Spotify Dataset (2000–2019) to predict whether a song would achieve popular status, defined as a popularity score above 70.

Using this dataset, I applied machine learning classification models to analyze features like danceability, energy, tempo, acousticness, valence, and loudness. My goal was to identify which attributes most influence a song's popularity and create predictive models to classify songs as likely hits or not. Through exploratory data analysis (EDA), I examined trends in these musical features and their relationships with popularity. I also performed feature engineering and preprocessing, encoding categorical variables and creating a binary target variable (`is_popular`). The prepared data was then split into training and testing sets for modeling.

I trained four machine learning models: Decision Tree, Naive Bayes, K-Nearest Neighbors (KNN), and Logistic Regression. Accuracy scores ranged from 0.615 to 0.652, with Logistic Regression performing the best. I also created visualizations to illustrate patterns in features like danceability, explicit content, and their impact on popularity.

This project allowed me to apply practical data science skills, including data preprocessing, model training, evaluation, and visualization, while gaining insight into how musical attributes can predict global popularity. It provided a hands-on opportunity to explore predictive modeling from my own perspective, combining analytical thinking with real-world data applications.



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III. IMPORTING LIBRARIES AND LOADING DATA

I began by importing the required tools (Pandas, Matplotlib) and loading our Spotify_Data.csv dataset.

```
import pandas as pd
import numpy
as np
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

df = pd.read_csv('Spotify_Data.csv')
df.head()
```

	artist	song	duration_ms	explicit	year	popularity	danceability	energy	key	loudness	mode	speechiness	acousticness	instrumentalness	liveness	valence	tempo	genre
0	Britney Spears	Oops!...I Did It Again	211160	False	2000	77	0.751	0.834	1	-5.444	0	0.0437	0.3000	0.000018	0.3550	0.894	95.053	pop
1	blink-182	All The Small Things	167066	False	1999	79	0.434	0.897	0	-4.918	1	0.0488	0.0103	0.000000	0.6120	0.684	148.726	rock, pop
2	Faith Hill	Breathe	250546	False	1999	66	0.529	0.496	7	-9.007	1	0.0290	0.1730	0.000000	0.2510	0.278	136.859	pop, country
3	Bon Jovi	It's My Life	224493	False	2000	78	0.551	0.913	0	-4.063	0	0.0466	0.0263	0.000013	0.3470	0.544	119.992	rock, metal
4	*NSYNC	Bye Bye Bye	200560	False	2000	65	0.614	0.928	8	-4.806	0	0.0516	0.0408	0.001040	0.0845	0.879	172.656	pop



IV. DATA INFORMATION

Here I identified that there are 2000 records of tracking data measuring features such as Acousticness, Danceability, Energy, Valence, Tempo, and overall Popularity.

```
print(df.info())  
print("\nDataset Shape:", df.shape)
```

```
<class 'pandas.DataFrame'>  
RangeIndex: 2000 entries, 0 to 1999  
Data columns (total 18 columns):  
#   Column              Non-Null Count  Dtype  
---  ---                -  
Dtype 0 0 artist  
-----  
2000 non-null str  
1 song 2000 non-null str  
2 duration_ms 2000 non-null int64  
3 explicit 2000 non-null bool  
4 year 2000 non-null int64  
5 popularity 2000 non-null int64  
6 danceability 2000 non-null float64  
7 energy 2000 non-null float64
```



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8	key	2000 non-null
	int64	
9	loudness	2000 non-null
	float64	
10	mode	2000 non-null
	int64	
11	speechiness	2000 non-null
	float64	
12	acousticness	2000 non-null
	float64	
13	instrumentalness	2000 non-null
	float64	
14	liveness	2000 non-null
	float64	
15	valence	2000 non-null
	float64	
16	tempo	2000 non-null
	float64	
17	genre	2000 non-null
	str	

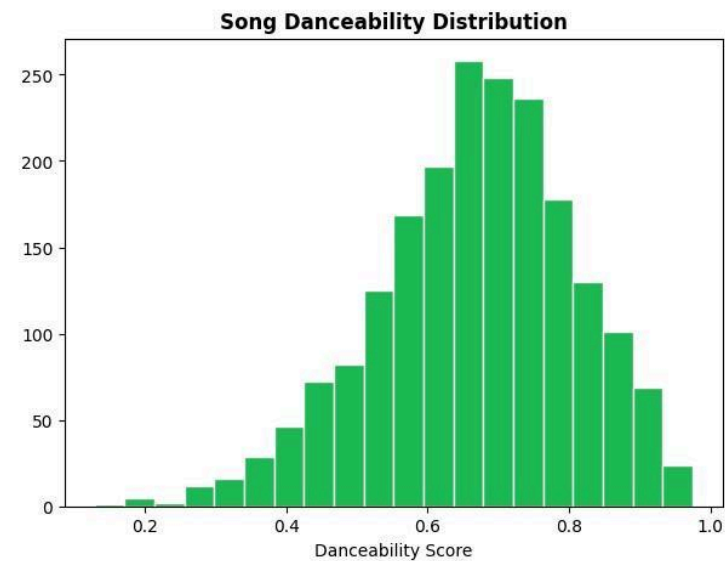
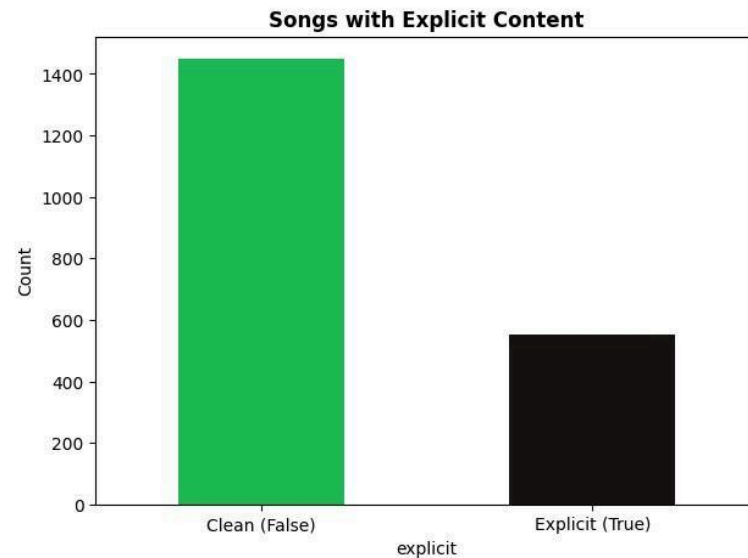
dtypes: bool(1), float64(9),
int64(5), str(3) memory usage: 267.7
KB None

Dataset Shape: (2000, 18)

V. EXPLORATORY DATA ANALYSIS

This is where I figured out how Danceability and Explicit Language map to overall Song Popularity using Matplotlib.

```
# Explicit Content Distribution plt.figure(figsize=(7, 5))
df['explicit'].value_counts().plot(kind='bar', color=['#1DB954',
'#191414']) plt.title("Songs with Explicit Content", fontweight='bold')
plt.ylabel("Count") plt.xticks(ticks=[0, 1], labels=["Clean (False)",
"Explicit (True)"], rotation=0) plt.show()
# Danceability Histogram plt.figure(figsize=(7, 5))
plt.hist(df['danceability'], bins=20, color='#1DB954',
edgecolor='white') plt.title("Song Danceability Distribution",
fontweight='bold') plt.xlabel("Danceability Score") plt.show()
```





VI. DATA PREPROCESSING AND FEATURE EXTRACTION

Prior to model training, I prepared the dataset by removing string identifiers (like Artist Name and Song Name). Crucially, I formatted the Target Variable y as a binary flag `is_popular` (1 for Popularity > 70, else 0).

```
# Drop textual details
df = df.drop(['artist', 'song'], axis=1)

# Encode categorical (genres and bools) from sklearn.preprocessing import
LabelEncoder le = LabelEncoder() for col in
df.select_dtypes(include=['object', 'string', 'bool']).columns:

    df[col] = le.fit_transform(df[col])

# Create Target Variable
df['is_popular'] = df['popularity'].apply(lambda x: 1 if x > 70 else 0) df
= df.drop('popularity', axis=1)

# Define X and Y
X = df.drop('is_popular', axis=1)
y = df['is_popular']

# Train Test Split
from sklearn.model_selection import train_test_split
X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)
print("Training Set Size:", len(X_train)) print("Testing Set Size:", len(X_test))
```

Training Set Size: 1600

Testing Set Size: 400



VII. TRAINING THE MACHINE LEARNING MODELS

I trained four Machine Learning classifiers on our Spotify Features (X) to accurately predict whether a track hits global popularity statuses (y).

```
from sklearn.tree import DecisionTreeClassifier from
sklearn.naive_bayes import GaussianNB from
sklearn.neighbors import KNeighborsClassifier from
sklearn.linear_model import LogisticRegression from
sklearn.metrics import accuracy_score

# 1. Decision Tree
dt = DecisionTreeClassifier(random_state=42) dt.fit(X_train,
y_train)
dt_acc = accuracy_score(y_test, dt.predict(X_test))

# 2. Naive Bayes nb =
GaussianNB()
nb.fit(X_train, y_train)
nb_acc = accuracy_score(y_test, nb.predict(X_test))

# 3. KNN
knn = KNeighborsClassifier(n_neighbors=5)
knn.fit(X_train, y_train) knn_acc =
accuracy_score(y_test, knn.predict(X_test))

# 4. Logistic Regression lr =
LogisticRegression(max_iter=2000, random_state=42)
lr.fit(X_train, y_train) lr_acc =
accuracy_score(y_test, lr.predict(X_test))

print("Decision Tree Accuracy:", round(dt_acc, 3)) print("Decision Tree Accuracy:", round(dt_acc, 3))
print("Naive Bayes Accuracy:", round(nb_acc, 3)) print("Naive Bayes Accuracy:", round(nb_acc, 3))
```

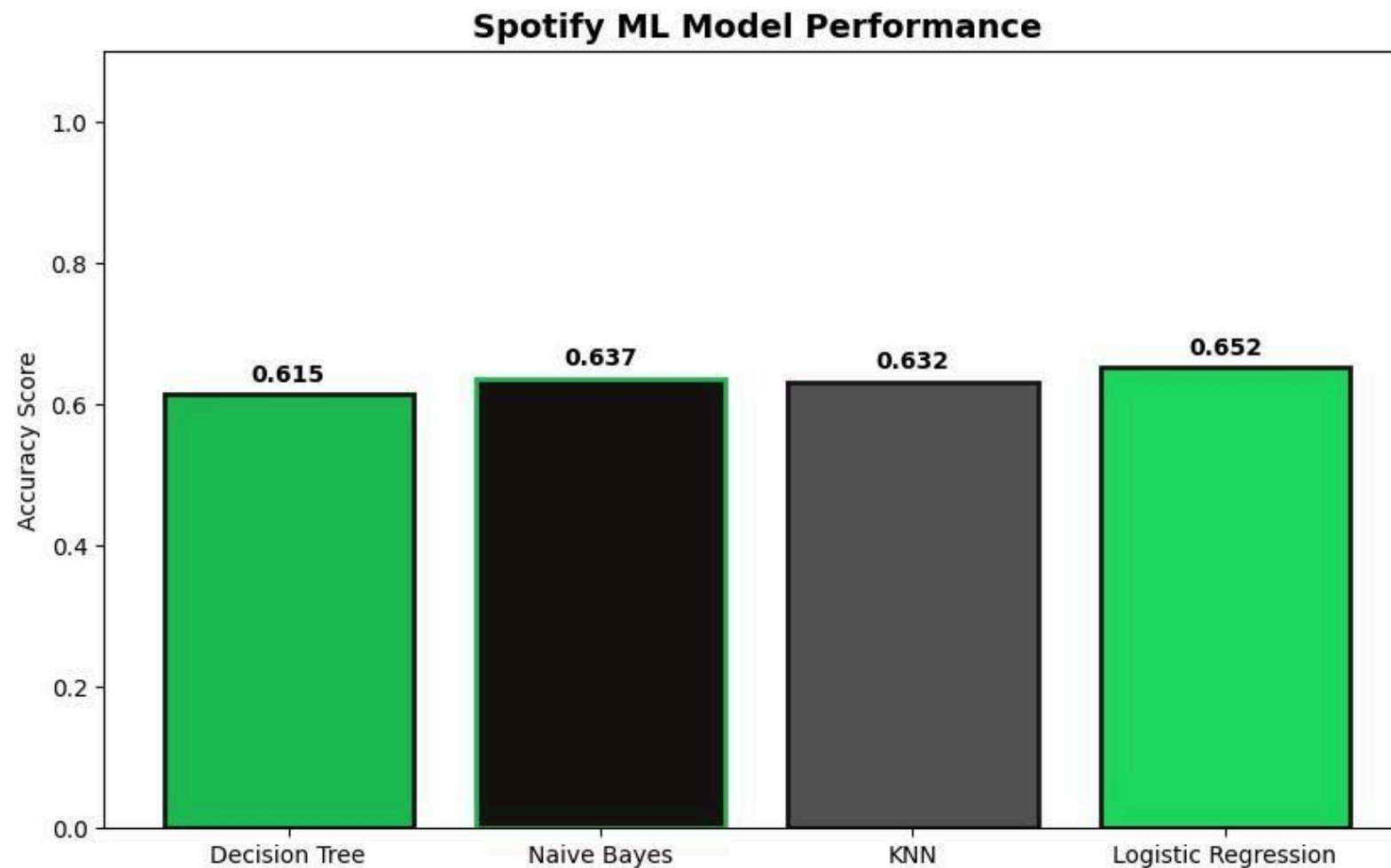


```
print("KNN Accuracy:", round(knn_acc, 3)) print("KNN Accuracy:", round(knn_acc, 3))  
print("Logistic Regression Accuracy:", round(lr_acc, 3)) print("Logistic Regression Accuracy:", round(lr_acc, 3))
```

Decision Tree Accuracy: 0.615
Naive Bayes Accuracy: 0.637
KNN Accuracy: 0.632
Logistic Regression Accuracy: 0.652

Model Accuracy Comparison Visualization

```
models = ["Decision Tree", "Naive Bayes", "KNN", "Logistic Regression"] scores  
= [dt_acc, nb_acc, knn_acc, lr_acc]  
  
plt.figure(figsize=(10, 6)) colors = ['#1DB954', '#191414', '#535353',  
'#1ED760'] # Spotify Theme Colors edges = ['#191414', '#1DB954', '#191414',  
'#191414']  
bars = plt.bar(models, scores, color=colors, edgecolor=edges, linewidth=2)  
  
for bar in bars:  
    yval = bar.get_height()  
    plt.text(bar.get_x() + bar.get_width()/2, yval + 0.01, f'{yval:.3f}', ha='center', va='bottom', fontweight='bold')  
  
plt.title("Spotify ML Model Performance", fontsize=14, fontweight='bold')  
plt.ylabel("Accuracy Score") plt.ylim(0, 1.1) plt.show()
```





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VIII. CONCLUSION

The analysis of 2000 Spotify tracks demonstrated that machine learning models could identify structural patterns in song attributes—such as danceability, tempo, and acousticness—that influence global popularity. Logistic Regression performed best in classifying songs, highlighting the predictive potential of musical features. This project provided practical experience in **data visualization, feature engineering, and predictive modeling**, allowing me to explore song popularity trends from a data-driven perspective.